5 In 2019 scientists developed a model for comparing the ages of humans and dogs. According to the model,

$Y = A\ln X + B$

where X = dog age in years and Y = human age in years.

For the model, it is known that when $X = 1$, $Y = 31$ and when $X = 12$, $Y = 71$.

(a) Find the value of B. **[1]**

(b) Determine the value of A, correct to the nearest whole number. **[2]**

Use the model, with the exact value of B and the value of A correct to the nearest whole number, to answer parts **(c)** and **(d)**.

(c) Find the human age corresponding to a dog age of 20 years. **[1]**

(d) Determine the dog age corresponding to a human age of 120 years. **[3]**

5(a)	
5(b)	

5(c)

5(d)